STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-CHARACTERIZE-GRADED-INJECTIVES-IN-DG

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3794-3816 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 3818-3848, the complete author proof (29 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 29 of queue_sdga.json (pb=914, stmt 3794-3816, proof 3818-3848).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

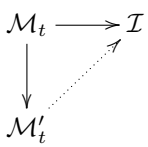
Source reference. Original lemma 0FSN: sources/stacks-sdga/pinned-original-source.tex, source lines 3696–3710. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSK: sources/stacks-sdga/pinned-original-source.tex, source lines 3513–3520. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSL: sources/stacks-sdga/pinned-original-source.tex, source lines 3573–3582. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{A}, d)$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. There exists a set T and for each $t \in T$ an injective map $\mathcal{M}_t \to \mathcal{M}'_t$ of acyclic differential graded $\mathcal{A}$-modules such that for an object $\mathcal{I}$ of $Mod(\mathcal{A}, d)$ the following are equivalent*

(1) *$\mathcal{I}$ is graded injective, and*
(2) *for every solid diagram*



Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

a dotted arrow exists in $Mod(\mathcal{A}, d)$ making the diagram commute.

Proof. Let T and $\mathcal{N}_t \to \mathcal{N}'_t$ be as in Lemma 0FSN. Denote $F : Mod(\mathcal{A}, \mathrm{d}) \to Mod(\mathcal{A})$ the forgetful functor. Let G be the left adjoint functor to F as in Lemma 0FSK. Set

$$\mathcal{M}_t = G(\mathcal{N}_t) \to G(\mathcal{N}'_t) = \mathcal{M}'_t$$

This is an injective map of acyclic differential graded $\mathcal{A}$-modules by Lemma 0FSL. Since G is the left adjoint to F we see that there exists a dotted arrow in the diagram

$$\begin{array}{ccc} \mathcal{M}_t & \longrightarrow & \mathcal{I} \\ \downarrow & \nearrow & \\ \mathcal{M}'_t & & \end{array}$$

if and only if there exists a dotted arrow in the diagram

$$\begin{array}{ccc} \mathcal{N}_t & \longrightarrow & F(\mathcal{I}) \\ \downarrow & \nearrow & \\ \mathcal{N}'_t & & \end{array}$$

Hence the result follows from the choice of our collection of arrows $\mathcal{N}_t \to \mathcal{N}'_t$. $\square$